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RESEARCH ARTICLE

Whole exome sequencing analyses identified novel genes for Alzheimer's disease and related dementia

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Abstract

INTRODUCTION: The heritability of Alzheimer's disease (AD) is estimated to be 58%–79%. However, known genes can only partially explain the heritability.

METHODS: Here, we conducted gene-based exome-wide association study (ExWAS) of rare variants and single-variant ExWAS of common variants, utilizing data of 54,569 clinically diagnosed/proxy AD and related dementia (ADRD) and 295,421 controls from the UK Biobank.

RESULTS: Gene-based ExWAS identified 11 genes predicting a higher ADRD risk, including five novel ones, namely *FRMD8*, *DDX1*, *DNMT3L*, *MORC1*, and *TGM2*, along with six previously reported ones, *SORL1*, *GRN*, *PSEN1*, *ABCA7*, *GBA*, and *ADAM10*. Single-variant ExWAS identified two ADRD-associated novel genes, *SLCO1C1* and *NDNF*. The identified genes were predominantly enriched in amyloid-β process pathways, microglia, and brain regions like hippocampus. The druggability evidence suggests that *DDX1*, *DNMT3L*, *TGM2*, *SLCO1C1*, and *NDNF* could be effective drug targets.

DISCUSSION: Our study contributes to the current body of evidence on the genetic etiology of ADRD.

KEYWORDS
Alzheimer's disease and related dementia, common variants, exome-wide association study, rare variants

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Highlights

- Gene-based analyses of rare variants identified five novel genes for Alzheimer's disease and related dementia (ADRD), including *FRMD8*, *DDX1*, *DNMT3L*, *MORC1*, and *TGM2*.
- Single-variant analyses of common variants identified two novel genes for ADRD, including *SLCO1C1* and *NDNF*.
- The identified genes were predominantly enriched in amyloid-β process pathways, microglia, and brain regions like hippocampus.
- *DDX1*, *DNMT3L*, *TGM2*, *SLCO1C1*, and *NDNF* could be effective drug targets.

1 | INTRODUCTION

Alzheimer's disease (AD), the most common type of dementia, possesses a high level of heritability estimated to be between 58% and 79%.[1] Characterizing the genetic architecture of AD would improve our understanding of pathological mechanisms and enable us to identify new therapeutic targets for AD. Previous genome-wide association studies (GWASs) have identified hundreds of AD-associated common variants in genes like *apolipoprotein E* (*APOE*), *BIN1*, and *CLU*.[2] However, these GWASs were limited by the coverage density of the genome and the power to detect low-frequency variants with strong effects.[3] Whole exome sequencing (WES) is effective in discovering causal and protein-coding rare variants,[4] which can fill the literature gap in the genetic architecture of AD. Utilizing WES, several rare variants in *TREM2*, *SORL1*, *ABCA7*, *ATP8B4*, and *ABCA1* have been highlighted for AD.[5] Nevertheless, the currently reported common and rare variants can only partially explain the heritability of AD. Thus, it is necessary to enlarge the WES sample to explore additional AD-related variants and corresponding genes.

In this study, we first performed the gene-based exome-wide association study (ExWAS) of rare variants (minor allele frequency [MAF] < 1%) and single-variant ExWAS of common variants (MAF ≥ 1%) for AD and related dementia (ADRD) incorporating WES data of 349,990 participants from UK Biobank (UKB). The results were validated in other independent cohorts. Second, we explored the biological function of ExWASs-identified genes in phenotype, tissue, cell, pathway, and protein-protein interaction network (PPIN). Third, we tested the causal and longitudinal relationships of ExWASs-identified genes with ADRD. Finally, for the identified novel genes, we looked up their potential in drug development (Figure 1). We aim to decipher the genetic basis and underlying pathological mechanism of ADRD.

2 | METHODS

2.1 | Study population

We included participants with WES data of interim 450k release from UKB (https://www.ukbiobank.ac.uk/), a population-based prospective cohort that enrolled half a million participants aged 40–69 years at baseline between 2006 and 2010.[6] UKB received ethical approval from the North West Multi-centre Research Ethics Committee and obtained written informed consent from all participants. Extensive genotypic and phenotypic data were collected from questionnaires, physical measures, sample essays, accelerometry, imaging, genotyping resources, and linked national health datasets during 10 years of follow-up. This study was conducted under application number 19,542.

2.2 | Phenotype definition

To maximize our ability to discover novel genetic associations, we used a definition of clinically diagnosed/proxy ADRD according to previous studies.[7] In detail, clinically diagnosed ADRD cases were participants diagnosed with AD or dementia according to the International Classification of Diseases (ICD)–10 codes (F00, F01, F02, F03, and G30) using information from health records (Categories 2405, 2406, 47, and 713) and Death Register data (Category 100093). Proxy ADRD cases were participants who reported the family history of AD or dementia of parents or siblings in the questionnaire (Fields 20107, 20110, and 20111).

2.3 | Exome sequencing and quality control

WES for all participants was conducted in the Regeneron Genetics Center (RGC) and the detailed methods have been described by members of the RGC.[8,9] Briefly, exomes were captured using the IDTxGen Exome Research Panel, with the initial 50,000 samples processed with IDT "lot1" and all others with "lot2." Illumina NovaSeq 6000 platform was used to sequence the samples with dual-indexed 75×75 bp paired-end reads using S2 and S4 flow cells for the initial and subsequent samples respectively, which achieved more than 20x coverage over 95% of the targeted bases.

In this study, we utilized the OQFE WES pVCF files in the GRCh38 human reference genome build[10] provided by UKB (Category 170) and performed additional quality control consistent with the previous study.[11] For variants, we utilized Hail to perform the genotype quality control. Multi-allelic sites were split into bi-allelic sites and all calls with low genotype quality and extremely low/high genotype depth were set to no call. We also removed variants with call rate < 90% or Hardy-Weinberg equilibrium (HWE) *p*-value $< 1.0 \times 10^{-15}$ and variants in Ensembl low-complexity regions. For participants, duplicate samples, samples withdrawn from the study, samples with discordance between self-reported and genetically inferred sex, and samples with transitions/transversions (Ti/Tv) ratio, heterozygous/homozygous (Het/Hom) ratio, single nucleotide variant/insertion-deletion (SNV/Indel) ratio or singletons number exceeding eight standard deviations from the mean values were removed. Further, we kept an unrelated sample using the kinship coefficient threshold at 0.0884, which denoted the second relatedness. King software was used to calculate pairwise heterozygote concordance rates and the kinship coefficient using the high-quality variants (MAF > 0.1%, missingness < 1%, HWE *p*-value $> 1.0 \times 10^{-6}$, and two rounds of pruning using –indep-pairwise 200 100 0.1 and –indep-pairwise 200 100 0.05). To maximize the sample size, for pairs with kinship coefficient > 0.0884, we first iteratively removed samples related to multiple other individuals until none remained, and then we removed one of the remaining pairs at random. Additionally, we restricted the sample to White British (Filed 22006) and calculated within-ancestral principal components (PCs) using the high-quality variants. Finally, a total of 349,990 participants and 20,080,554 variants passed quality control and were included in the main analyses.

2.4 | Variant annotation

Rare variants (MAF < 1%) that successfully passed quality control procedures were subjected to further annotation using VEP version 108.[12] Loss-Of-Function Transcript Effect Estimator (LOFTEE) was employed to identify variants associated with a substantial burden of natural loss-of-function (LOF), encompassing stop-gained, essential splice, and frameshift variants.[13] Only those variants predicted with high confidence were retained for subsequent analysis. Rare Exome Variant Ensemble Learner (REVEL) score was utilized to quantify the selective pressure exerted against missense variants in each gene. Two potential thresholds were used (REVEL score of 50 and 75, score range 0–100) to define a variant to be pathogenic as recommended by a previous study.[14] Based on these annotations, variants were classified into three distinct consequence categories: LOF, REVEL scores above 75, and between 50 and 75. To continuously measure the selective pressure against mutations, two more aggregated thresholds were established: LOF combined with REVEL ≥75, and LOF combined with REVEL ≥50. For the gene-based collapsing test, variants within each gene were aggregated accordingly.

Common variants (MAF ≥ 1%) were annotated with ANNOVAR[15] using refGene (https://annovar.openbioinformatics.org/en/latest/user-guide/download/) as a reference panel. Variants annotated as exonic, UTR3, or UTR5 were retained in the following single-variant ExWAS analyses.

RESEARCH IN CONTEXT

1. **Systematic review**: Previous genome-wide association studies of Alzheimer's disease (AD) were mainly based on genotyped array data of common variants, which reported plenty of AD-associated genes with low effect sizes. Few genetic studies were performed using whole-genome sequencing data of rare variants, and the sample sizes of these studies were relatively small.
2. **Interpretation**: Here, we conducted exome-wide association study of AD and related dementia (ADRD) utilizing data of 349,990 participants from the UK Biobank. We validated the genes from previous literature, and more importantly, we identified seven novel genes for ADRD, including *FRMD8*, *DDX1*, *DNMT3L*, *MORC1*, *TGM2*, *SLCO1C1*, and *NDNF*. This study added new evidence on the genetic etiology of ADRD.
3. **Future directions**: Future research is necessary to confirm our findings in cohorts with different settings. In addition, basic studies are needed to explore the potential mechanisms of the identified genes to provide perspectives for therapy.

2.5 | Gene-based ExWAS

Gene-based ExWAS, the collapsing test, was performed to identify rare (MAF < 1%) variants associated with ADRD. To investigate how rare variants with different pathogenicity and frequency contribute to the risk of dementia, multiple masks were implied in this study: 75 > REVEL > 50, REVEL > 75, LOF, LOF + REVEL ≥ 75, LOF + REVEL ≥ 50. Four frequency masks were considered for each of these masks: MAF < 0.001%, < 0.01%, < 0.1%, and < 1%. Thus, 20 gene-burden models were constructed for each gene and multiple corrections were conducted within all gene-burden models. Since a recent study showed that adjustment for the polygenic risk score (PRS) of common variants can efficiently improve the power to detect rare variant associations,[16] standard PRS for AD was acquired[17] (Field 26206, detailed information can be found in https://biobank.ndph.ox.ac.uk/showcase/refer.cgi?id = 5202). Gene-based ExWAS was performed using a generalized mixed model implemented in SAIGE-GENE+,[18] adjusting for age, sex, the first 10 PCs, and PRS. The sparse genetic relationship matrix used to estimate the model's variance ratio was constructed using high-quality variants with the recommended relative coefficient cutoff of 0.05. Leave-one-variant-out (LOVO) analyses were performed to assess the stability of the results and to identify the most important variant driving the significant associations in the collapsing test, that is, the variant that maximized the alteration of the *p*-value after removing it. Gene-based collapsing tests in subgroups stratified by sex (female and male), age of onset (early-onset [< 65 years] and late-onset [≥ 65 years]), and *apolipoprotein E (APOE) ε4* carrier status (non-carrier

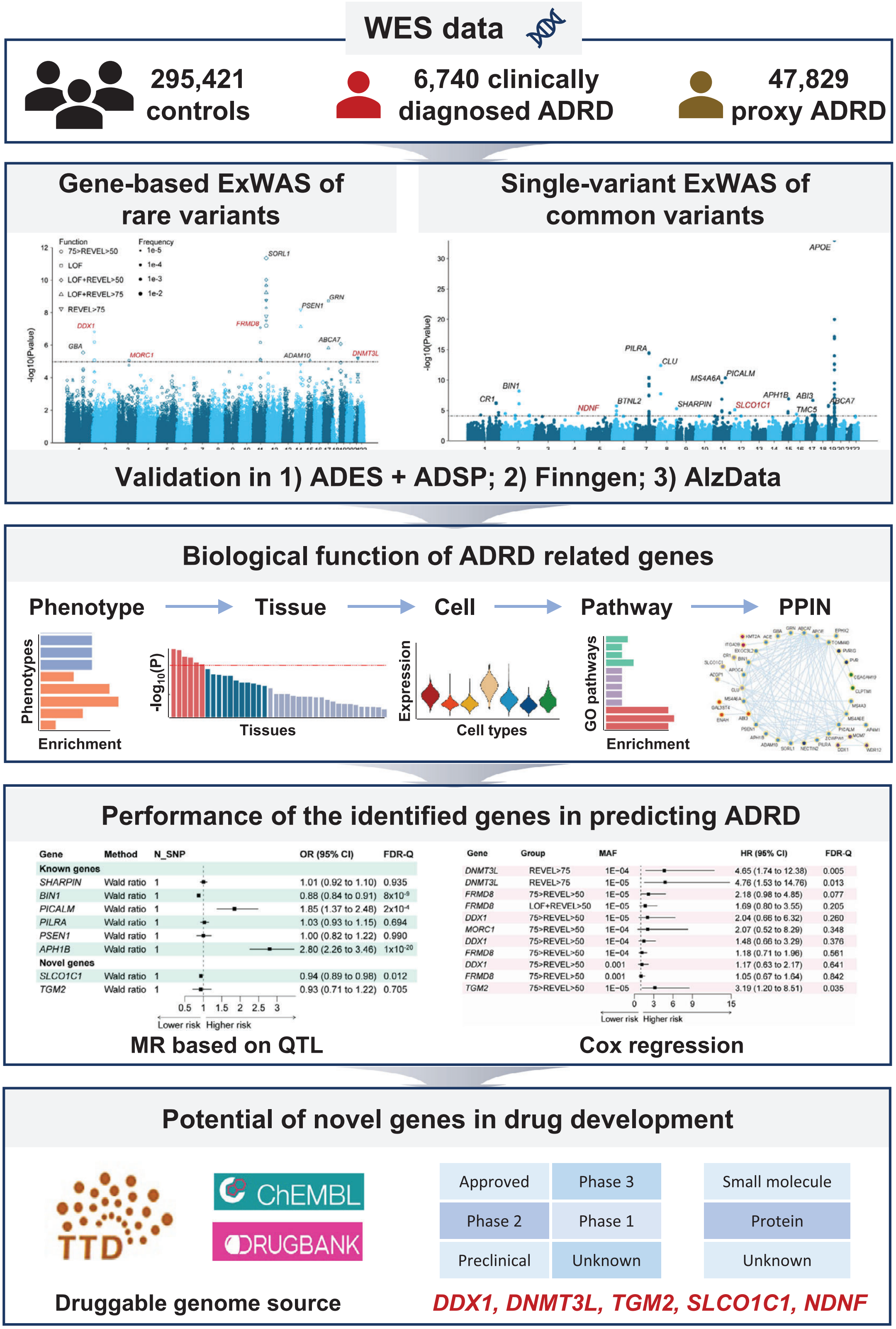

FIGURE 1 Study workflow. Incorporating the WES data from UKB, we first performed the gene-based ExWAS of rare variants and single-variant ExWAS of common variants for ADRD. Second, we explored the biological function of ExWASs identified genes in phenotype, tissue, cell, pathway, and PPIN. Third, we explored the causality and predicting performance of the identified genes in ADRD. Finally, for the robust novel genes, we investigated their potential in drug development.

and carrier, defined by rs429358 and rs7412) were performed to investigate the heterogeneity of ADRD-associated genes in different populations. For the identified significant genes, we further tested their specificity in clinically diagnosed ADRD, proxy ADRD, AD, and vascular dementia (VD). In our collapsing analysis, we mainly reported the *p*-value of the SKAT-O test. Since multiple masks contained identical rare variants for a part of genes, we removed duplicated results and performed the Benjamini-Hochberg false discovery rate (FDR) correction, with FDR-Q < 0.05 considered significant as previous studies.[5,11,19]

To assess the interaction of rare and common variants, we conducted conditional analyses to figure out whether the nearby common variants (located within ± 500 kb of the identified gene) influence the effect of genes identified from gene-based ExWAS on ADRD. We first performed GWAS using UKB genotype data (Category 263) to search for nearby ADRD-related common variants. Detailed methodological description of the detection, imputation, and quality control of UKB genotyping data can be found in previous study.[20] Briefly, IMPUTE4 software was used to perform the imputation with the Haplotype Reference Consortium as a reference panel. We additionally removed the variants with call rate < 90%, MAF < 0.005, HWE *p*-value < 1×10^{-6}, and imputation quality score < 0.5. Samples with missing rate > 0.05, sex mismatches, abnormal sex chromosome aneuploidy, heterozygosity rate outliers, those who were not white British, and those who had more than 10 putative third-degree relatives were further excluded. After clumping (–clump-p1 1×10^{-5} –clump-r^2 0.01), we obtained the independent ADRD-related common variants and then repeated the collapsing tests by adding these variants as covariates. Besides, we further tested the aggregate effect of the identified genes and PRS on ADRD. To evaluate the influence of *APOE*, both the PRS containing the *APOE* region and excluding the *APOE* region were calculated following the same procedures in our previous work.[21] Using similar quality control parameters for imputed SNP genotype data in the above conditional analysis (imputation quality score < 0.8), we performed the classic clumping and thresholding method to generate PRS by PRSice2[22] with a *p*-value threshold of 5×10^{-8}. GWAS summary data from the International Genomics of Alzheimer's Project (IGAP)[23] were used as the discovery data. We divided the participants into 18 groups according to the carrier status (no vs. yes) of any rare variant of 10 identified genes, and PRS categories (0%–10%, 10%–20%, 20%–30%, 30%–40%, 40%–60%, 60%–70%, 70%–80%, 80%–90%, and 90%–100%). Combined (any of the identified genes & PRS) effects on ADRD were investigated using logistic regression taking "carrier status no & PRS category 40%–60%" as the reference group.

2.6 | Single-variant ExWAS

Single-variant ExWAS, the logistic regression, was performed to identify common (MAF ≥ 1%) variants associated with ADRD. Logistic regression was performed using PLINK v2,[24] adjusting for the age, sex, and the first 10 PCs. To determine common genetic loci related to ADRD, we used the UKB White British as the reference panel. Independent significant single nucleotide polymorphisms (SNPs) were identified by FDR < 0.05 and by their independence ($r^2 \leq 0.6$ within a 1 Mb window). Lead SNPs were defined as independently significant SNPs with $r^2 \leq 0.1$ within a 1 Mb window, and genomic loci were then defined by merging lead SNPs in 250 kb. Candidate SNPs were defined as those in linkage disequilibrium (LD) of one of the independent significant SNPs. Since proxy cases were also analyzed in our study, corrected effect sizes and standard errors[25] were also reported. The carrier status of *APOEε4*, the strongest susceptibility gene of AD, was additionally adjusted to investigate the effects of ADRD-associated variants independent of *APOE*. Similarly, subgroup single-variant association analyses stratified by sex, age of onset, and *APOEε4* carrier status were performed, and an exome-wide significance level of FDR-Q value 0.05 was set in the analyses. The specificity of the identified significant genes in clinically diagnosed ADRD, proxy ADRD, AD, and VD was also tested. All tests from different subgroups were included in the FDR correction simultaneously.

2.7 | Replication

We performed additional replication analyses, using summary statistics from another two independent cohorts. One of the cohorts was from the Alzheimer Disease European Sequencing (ADES) consortium and the Alzheimer's Disease Sequencing Project (ADSP)[26] (12,652 AD cases and 8693 controls), and the other was from the Finngen study[27] (7759 clinically diagnosed AD and 334,740 controls). Summary statistics of the ADES and ADSP were acquired from the stage 1 analyses from Holstege et al.[5] The diagnosis of AD in the ADES and ADSP was based on the National Institute on Aging-Alzheimer's Association (NIA-AA) criteria[28] or the National Institute of Neurological and Communicative Disorders and Stroke and the Alzheimer's Disease and Related Disorders Association (NINCDS-ADRDA) criteria.[29] Ordinal logistic regression was used to perform the burden test. Summary statistics of the Finngen were acquired from https://r8.finngen.fi/. The diagnosis of AD in the Finngen was based on ICD-9 and ICD-10 codes, and the summary data of G6_ALZHEIMER were acquired in this study. Finngen ran association tests with regenie,[30] and the model was corrected for age, sex, batch, and the first 10 principal components. For the gene identified in the gene-based ExWAS and stratification analyses, a minimal *p*-value was kept for each gene in the summary statistics. As for the significant associations identified in the single-variant ExWAS, the associations with the lead variants were queried. A gene or variant was considered replicated if the associations reached $p < 0.05$.

2.8 | Analysis of publicly available expression datasets from NCBI GEO

To investigate the associations between identified genes and AD, we performed a lookup in expression level between AD (*N* = 684) and control (N = 562) from human *post mortem* brain samples (http://www.alzdata.org/download1.php). The microarray and RNA-seq brain expression profiling data were integrated and normalized across different platforms by Xu et al.[31] Expression profiling data were subjected to data normalization and $\log_2$ transformation and were integrated using empirical Bayesian methods to reduce batch effects. A linear regression model was used to correct for age and sex to investigate whether the specific expression levels of ADRD-related genes identified from ExWAS differed in the entorhinal cortex (EC), hippocampus (HP), temporal cortex (TC), and frontal cortex (FC) between samples with and without AD. Benjamini-Hochberg FDR correction was used, with FDR-Q < 0.05 considered significant.

2.9 | Phenotype, tissue, and pathway enrichment analyses

We explored the phenotype specificity, tissue enrichment, and pathway enrichment of ADRD-associated genes identified from gene-based and single-variant ExWASs (including 10 genes from gene-based ExWASs and 52 genes mapped from candidate SNPs from single-variant ExWASs, a total of 61 genes) using the GENE2FUNC function in FUMA with all genes as background genes. Briefly, we performed a hypergeometric test to assess the enrichment of ADRD-associated genes in specific traits or diseases in the GWAS catalog.[32] To test for tissue enrichment of ADRD-associated genes, differentially expressed gene (DEG) sets were pre-calculated by performing a two-sided t-test for any tissue type against all other 54 tissue types in GTEx v8.[33] The hypergeometric test was then used to test ADRD-associated genes against each of the DEG sets. As for pathway enrichment, we performed a hypergeometric test to examine whether ADRD-associated genes were overrepresented in any of the predefined gene sets in FUMA. To gain some insights into the enriched pathways of specific genes, we also performed the enrichment analyses restricted to genes identified from ExWAS of rare variants and novel genes identified from ExWAS of rare variants.

2.10 | Single-nucleus RNA sequencing (snRNA-seq) data source and differential expression analyses

The snRNA-seq data of the human prefrontal cortex from late-stage AD patients (n = 11) and age-matched cognitively healthy controls (n = 7) were obtained from a recent study by Morabito et al. on Gene Expression Omnibus database with the accession ID: GSE174367.[34] Cell type annotation was conducted using the metadata file provided by the authors. The R package Seurat[35] was used for the main analyses and visualization of the results, where function AddModuleScore was used to calculate the gene set score as the relative expression level of the ADRD-associated genes identified from ExWASs (including 10 genes from gene-based ExWASs and 52 genes mapped from candidate SNPs from single-variant ExWASs, a total of 61 genes).

2.11 | PPIN analyses

PPIN analyses were performed using ADRD-associated genes identified from ExWASs (including 10 genes from gene-based ExWASs and 52 genes mapped from candidate SNPs from single-variant ExWASs, a total of 61 genes) as input into the STRING database (Version 12.0).[36] Using the default confidence score, only links with at least medium confidence were plotted. Markov clustering (MCL) was further performed to investigate the functional enrichments by clusters using the default settings.[37]

2.12 | Mendelian randomization (MR) analyses integrating quantitative trait loci (QTL) data

To validate the causal relationship between ADRD-associated genes identified from ExWASs and ADRD, we performed the MR analyses using the expression QTL (eQTL), splicing QTL (sQTL), and protein QTL (pQTL) as the exposure data. The eQTL data were obtained from MetaBrain,[38] which integrated 8613 RNA-sequencing samples from 14 brain datasets. In this study, the Cerebellum-EUR (n = 492) and Cortex-EUR (n = 2683) datasets were treated as exposure data. The pQTL data were obtained from brain regions of the dorsolateral prefrontal cortex (dPFC) and frontal cortex for ROS/MAP, dPFC for Banner, and parahippocampal gyrus for MSBB (n = 722).[39] Detailed information for proteomic data generation procedures can be found in previous papers.[40–42] The sQTL data were obtained from 2865 brain cortex samples from 2443 unrelated individuals of European ancestry.[43] Only cis-QTL data were used in this study. A GWAS[44] with 111,326 clinically diagnosed/proxy ADRD and 677,663 controls was used as the outcome data. Genome-wide significant (p-value $< 5\times10^{-8}$) SNPs were selected and LD clumping was performed to obtain independent SNPs ($r^2 > 0.001$). Exposure QTL and outcome (ADRD GWAS) data were then harmonized for the same effect alleles. When only a single independent QTL was available, the Wald ratio was used to estimate the causality from exposure to outcome. When more than one SNP was available, the inverse-variance weighted (IVW) method was used to combine the ratios of SNP-exposure to SNP-outcome in a fixed-effects or random-effects meta-analysis. The MR analyses were performed using the R package "TwoSampleMR" version 0.5.6.

2.13 | Cox proportional hazard regression analyses

The Cox proportional hazard regression model was used to estimate the longitudinal efficacy of carrying variants of ADRD-associated genes identified from ExWASs in predicting incident ADRD. Time to ADRD event was calculated from the date of birth (Field 33) until the date of earliest incident ADRD diagnosis, death, the last date of data collection by the general practitioner, or the last date of available hospital admission data, whichever came first. Sex, age, and the first

10 PCs were adjusted in the Cox model. For the *TGM2*, which was identified from stratification analysis in the *APOEε4* carrier subgroup, survival analysis was conducted in both the corresponding subgroup and the whole population. An additive effect of *TGM2* and *APOEε4* was also conducted. In each Cox regression analysis, the prognostic potential of the models was quantified using the concordance index (C-index),[45] derived from the result of the Coxph function. The models exhibited a moderate predictive capacity, with an average C-index of approximately 0.82.

2.14 | Druggability

To examine the druggability of ADRD-related genes, we searched the druggable genome source,[46] DrugBank database[47] (https://go.drugbank.com/), ChEMBL database[48] (https://www.ebi.ac.uk/chembl/), and Therapeutic Target Database (TTD)[49] (https://db.idrblab.net/ttd/) to find the potential drug targets for ADRD. All available information involving small molecules, proteins, and other drugs in any stage of clinical trial or preclinical phase was collected.

3 | RESULTS

3.1 | ExWASs identified ADRD-related genes

After quality control, we included 349,990 White British participants in the main analyses, including 54,569 clinically diagnosed/proxy ADRD (6740 clinically diagnosed ADRD and 47,829 proxy ADRD) and 295,421 controls (Table S1). The included participants had a mean age of 56.94 years at study enrolment, and 53.76% were female. A total of 1,209,264 rare variants were annotated as LOF variants or missense variants with REVEL scores exceeding 50, which were considered potentially pathogenic. As for common variants, a total of 50,506 variants were included in association tests.

LOF and missense rare variants were annotated for gene-based ExWAS utilizing LOFTEE and REVEL score, respectively. We defined up to 20 groups for each of 16,411 genes in collapsing tests according to the function (LOF, REVEL > 75, 75 > REVEL > 50, LOF + REVEL > 75, LOF + REVEL > 50) and frequency (MAF < 0.001%, < 0.01%, < 0.1%, and < 1%) of annotated variants. Among 137,140 gene-based collapsing tests for ADRD (Table S2), 28 associations covering 10 genes retained significance after FDR correction with FDR-Q < 0.05 (Figure 2A and Table 1), including 4 novel genes (*FRMD8*, *DDX1*, *DNMT3L*, and *MORC1*) (Figure 3) and 6 previously reported ones (*SORL1*, *GRN*, *PSEN1*, *ABCA7*, *GBA*, and *ADAM10*). When adjusted for the *APOEε4* carrier status, most of the associations remained significant (FDR-Q < 0.05) except for the results of *DNMT3L* (Table S2). In stratification analyses, five genes remained significant (FDR-Q < 0.05): *SORL1* in female, late-onset ADRD, and non-carrier of *APOEε4* subgroups; *PSEN1* in female and non-carrier of *APOEε4* subgroups; *DDX1* in late-onset ADRD subgroup; *GRN* and *ABCA7* in non-carrier of *APOEε4* subgroup. Specifically, we found another novel gene, *TGM2*, significant in carrier of *APOEε4* subgroup (Table S3). Considering that the proxy cases may influence the reliability of the results, we separately performed the EWAS of clinically diagnosed ADRD and proxy ADRD. All significant genes identified from the main analyses were correlated to a higher risk of both clinically diagnosed and proxy ADRD, but the P-values for *DDX1*, *MORC1*, *ABCA7*, and *ADAM10* were only significant in proxy ADRD (Table S4). AD is the most common type of ADRD followed by VD. We found that most of the ADRD-related genes were significantly related to AD except for *FRMD8*, *MORC1*, and *GBA*, while only *FRMD8*, *PSEN1*, and *ADAM10* were significantly related to VD (Table S4). LOVO analyses of the above genes were performed to assess the stability of our results and to identify the important variants that drive the significant associations between the identified genes and ADRD. Most associations remained robust in LOVO analyses (Table S5 and Figure S1), except for the association of *SORL1* attaining the highest LOVO *p*-value of 0.006 (75 > REVEL > 50 and MAF < 1% group) after removing its most important variant, chr11:121625107:A:T (Figure S2).

To investigate the potential influence of common variants on the effects of the identified genes, we conducted conditional analyses. GWAS was performed to identify common variants near the genes identified from gene-based ExWAS. We found one common variant for *SORL1* (chr11:121564878:T:C), one for *PSEN1* (chr14:73510230:T:C), one for *ABCA7* (chr19:1059244:T:A), and three for *ADAM10* (chr15:58849576:T:C, chr15:58698575:CA:A, chr15:58592773:T:A), but none for *GRN*, *FRMD8*, *DDX1*, *GBA*, *DNMT3L*, *MORC1*, and *TGM2*. We repeated the gene-based ExWAS by conditioning on the nearby common variants. The P-values of *SORL1*, *PSEN1*, *ABCA7*, and *ADAM10* changed slightly (Table S6), suggesting that these genes were independent of the nearby common variants. Apart from nearby common variants, we also explored the combined effect of ten genes identified from gene-based ExWAS and PRS derived from genome-wide common variants data. Compared to individuals with no variants of identified genes and middle level of PRS score, the odds ratio (OR) for ADRD displayed a clear increasing trend among participants with either detrimental variants or higher PRS score (Figure 4). Individuals carrying rare variants of identified genes and in PRS category 90%–100% reached an OR of 2.10 (95%CI 1.90–2.31, $p = 1.98\times10^{-49}$), which was 1.25 times as large as those not carrying rare variants of identified genes in the same PRS category. Considering the prominent effect of *APOE* in PRS, we also calculated the PRS by excluding the *APOE* region. Although the OR dropped a little, the additive effect remained (Table S7).

Single-variant ExWAS identified 15 genes, clumped from 80 candidate variants, significantly associated with ADRD (FDR-Q < 0.05) from 50,506 logistic regression tests (Table S8), including 2 novel genes (*SLCO1C1* and *NDNF*), as well as 13 previously reported ones (*APOE*, *PILRA*, *CLU*, *PICALM*, *MS4A6A*, *BIN1*, *APH1B*, *ABI3*, *CR1*, *ABCA7*, *BTNL2*, *SHARPIN*, and *TMC5*) (Figure 2B and Table 2). Notably, both rare and common variants contributed to the role of *ABCA7* in ADRD. All genes mentioned above remained significant (FDR-Q < 0.05) after adjustment of the *APOEε4* carrier status, indicating that the effects

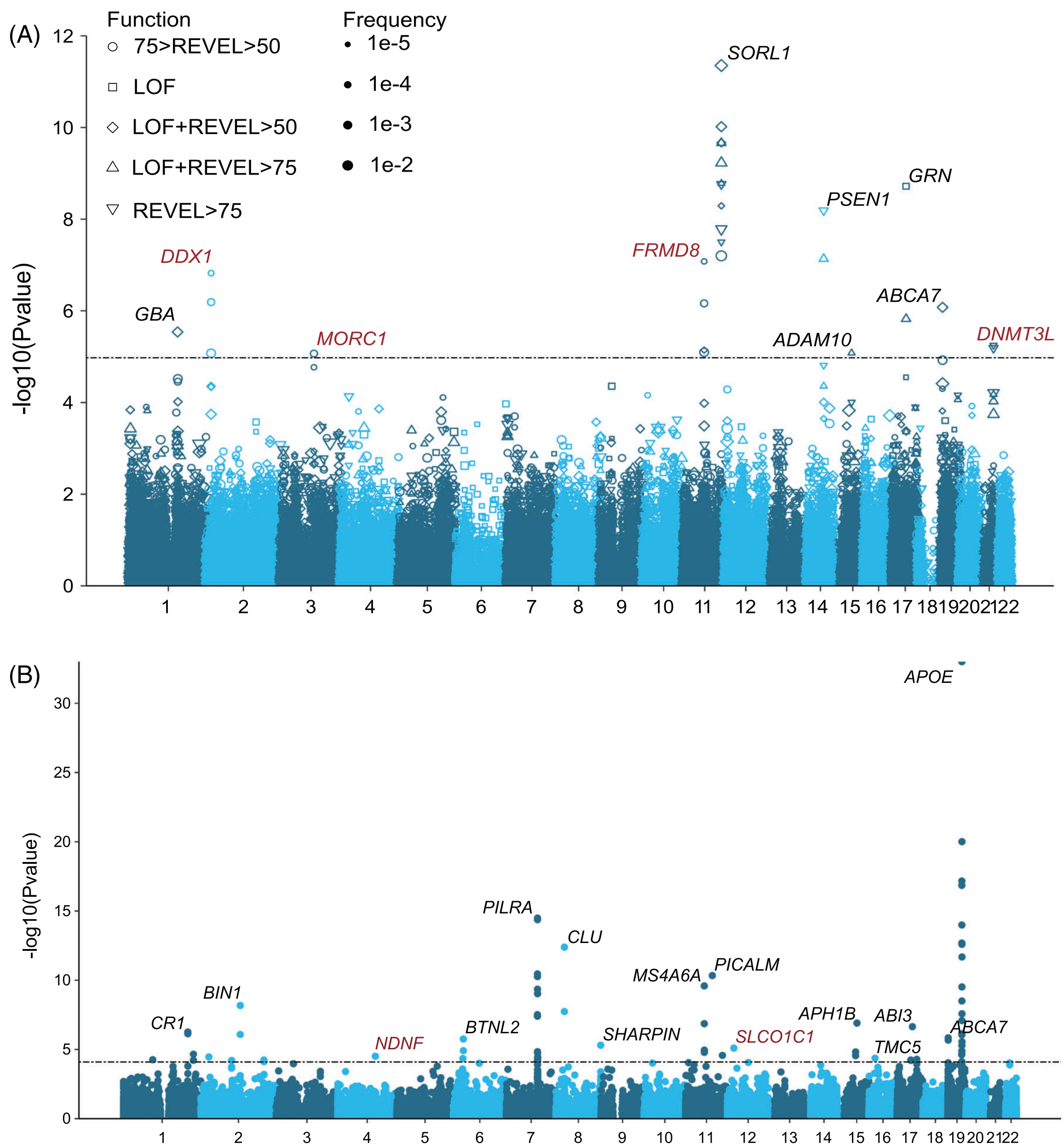

FIGURE 2 Manhattan plots of the gene-based (A) and single-variant (B) ExWASs for ADRD in UKB. The plot shows the -log10 of the *p*-value from against their genomic position for each genetic association with ADRD. The dotted line represents the significance level of $p = 1.05 \times 10^{-5}$ (A) and $p = 8.0 \times 10^{-5}$ (B). The identified genes containing the variant(s) that surpassed our significance threshold after FDR correction are marked above points, among which the novel ones are highlighted in red and the known ones in black.

of these ADRD-associated common variants were independent of *APOEε4* (Table S8). In stratification analyses, six genes clumped from 62 candidate variants were significantly (FDR-Q < 0.05) associated with ADRD (Table S9). Four genes (*PICALM*, *APOE*, *PILRA*, and *CLU*) have been identified in analyses of the entire population. Two additional genes were identified in subgroup analyses, including *EXOC3L2* (FDR-Q $= 8.7 \times 10^{-4}$) and *NECTIN2* (FDR-Q $= 4.1 \times 10^{-12}$), which were specifically associated with ADRD in the early-onset subgroup. When the proxy cases were excluded, the results remained significant except for *SLCO1C1* and *NDNF* (Table S10). Most of the ADRD-related genes were significantly related to AD except for *ABI3*, *NDNF*, and *TMC5*, while only *APOE*, *PILRA*, *CLU*, *PICALM*, and *ABI3* were significantly related to VD (Table S10).

3.2 | Validation of ADRD-related genes

To validate the robustness of the identified genes, we replicated the analyses in another two large genome-wide studies, including one from the ADES consortium and the ADSP,[5] and the other from the Finngen study.[27] Based on the summary statistics of the ADES consortium and ADSP, the deleterious roles of *SORL1*, *ABCA7*, and *ADAM10* in ADRD

TABLE 1 Significant (FDR-adjusted) results from gene-based ExWAS of ADRD in UKB.

Gene	Group	Max MAF	cMAC (case/control)	N_rare variants	OR	SE	*p*-value	FDR-Q
SORL1	LOF + REVEL > 50	0.01	5390 (1027/4363)	380	1.011	0.002	4.E-12	6.E-07
	LOF + REVEL > 50	0.001	2637 (502/2135)	379	1.010	0.002	1.E-10	7.E-06
	LOF + REVEL > 50	1.E-04	1692 (347/1345)	372	1.014	0.003	2.E-10	7.E-06
	LOF + REVEL > 75	1.E-04	581 (135/446)	173	1.022	0.004	2.E-10	7.E-06
	LOF + REVEL > 75	0.001	1152 (239/913)	176	1.016	0.003	6.E-10	2.E-05
	LOF + REVEL > 75	1.E-05	290 (84/206)	156	1.038	0.006	2.E-09	3.E-05
	REVEL > 75	1.E-04	540 (124/416)	149	1.021	0.005	2.E-09	3.E-05
	LOF + REVEL > 50	1.E-05	564 (140/424)	308	1.027	0.005	5.E-09	8.E-05
	REVEL > 75	0.001	1111 (228/883)	152	1.015	0.003	2.E-08	2.E-04
	REVEL > 75	1.E-05	258 (74/184)	133	1.037	0.007	3.E-08	4.E-04
	75 > REVEL > 50	0.01	4238 (788/3450)	204	1.010	0.002	6.E-08	7.E-04
GRN	LOF	1.E-04	114 (42/72)	23	1.059	0.010	2.E-09	3.E-05
	LOF + REVEL > 75	1.E-04	212 (58/154)	47	1.037	0.007	2.E-06	0.010
PSEN1	REVEL > 75	1.E-04	127 (43/84)	53	1.052	0.009	6.E-09	9.E-05
	LOF + REVEL > 75	1.E-04	228 (52/176)	62	1.025	0.008	7.E-08	7.E-04
FRMD8	75 > REVEL > 50	1.E-05	132 (45/87)	60	1.049	0.009	8.E-08	8.E-04
	75 > REVEL > 50	1.E-04	667 (118/549)	81	1.006	0.004	7.E-07	0.005
	LOF + REVEL > 50	1.E-05	198 (55/143)	93	1.035	0.008	7.E-06	0.041
	75 > REVEL > 50	0.001	964 (169/795)	83	1.006	0.004	8.E-06	0.042
DDX1	75 > REVEL > 50	1.E-05	77 (31/46)	37	1.061	0.011	1.E-07	0.001
	75 > REVEL > 50	1.E-04	212 (46/166)	46	1.017	0.007	6.E-07	0.005
	75 > REVEL > 50	0.001	449 (77/372)	48	1.005	0.005	8.E-06	0.042
ABCA7	LOF + REVEL > 50	0.001	6484 (1174/5310)	541	1.008	0.002	8.E-07	0.006
GBA	LOF + REVEL > 50	0.001	1832 (361/1471)	173	1.013	0.003	3.E-06	0.019
DNMT3L	REVEL > 75	1.E-05	41 (17/24)	16	1.076	0.016	6.E-06	0.035
	REVEL > 75	1.E-04	53 (18/35)	17	1.057	0.014	6.E-06	0.039
ADAM10	LOF + REVEL > 75	1.E-05	20 (11/9)	18	1.104	0.022	8.E-06	0.042
MORC1	75 > REVEL > 50	1.E-04	54 (20/34)	10	1.064	0.014	9.E-06	0.042

Abbreviations: cMAC, cumulative minor allele count; FDR, false discovery rate; LOF, loss-of-function; MAF, minor allele frequency; OR, odds ratio; SE, standard error.

were validated (Table S11 and Figure S3). When it comes to the results of the Finngen study, the associations of all genes identified from gene-based ExWAS with ADRD were significant ($p < 0.05$) including the novel genes *DDX1*, *DNMT3L*, *FRMD8*, *MORC1*, and *TGM2* (Table S12 and Figure S4). As for the genes identified from single-variant ExWAS, most of them were significant except for *BTNL2*, *SLCO1C1*, *NDNF*, and *TMC5* (Table S12 and Figure S4).

We further investigated the associations between identified genes and AD interrogating microarray and RNA-seq brain expression profiling data from AlzData.[31] DEG analysis was conducted between AD patients and healthy controls in critical brain regions of the EC, HP, TC, and FC. It is observed that 6 genes identified from gene-based ExWAS (*SORL1*, *GRN*, *PSEN1*, *ABCA7*, *DDX1*, and *FRMD8*), and 8 genes identified from single-variant ExWAS (*PILRA*, *CLU*, *MS4A6A*, *BIN1*, *APH1B*, *ABI3*, *ABCA7*, and *SHARPIN*) were differentially expressed in EC, HP, TC, or FC in AD patients versus healthy controls (Table S13).

3.3 | The biological function of ADRD-related genes

To explore the biological functions of genes identified in ExWAS, we investigated their specificity across phenotypes, tissues, cells, pathways, and PPIN. The top phenotypes enriched from the GWAS catalog were AD or family history of AD (Figure 5A and Table S14), further strengthening our results. Tissue specificity analyses demonstrated that these genes were differentially expressed, in 10 out of 54 tissue types (FDR < 0.05), including seven brain regions of the anterior cingulate cortex, nucleus accumbens, amygdala, caudate,

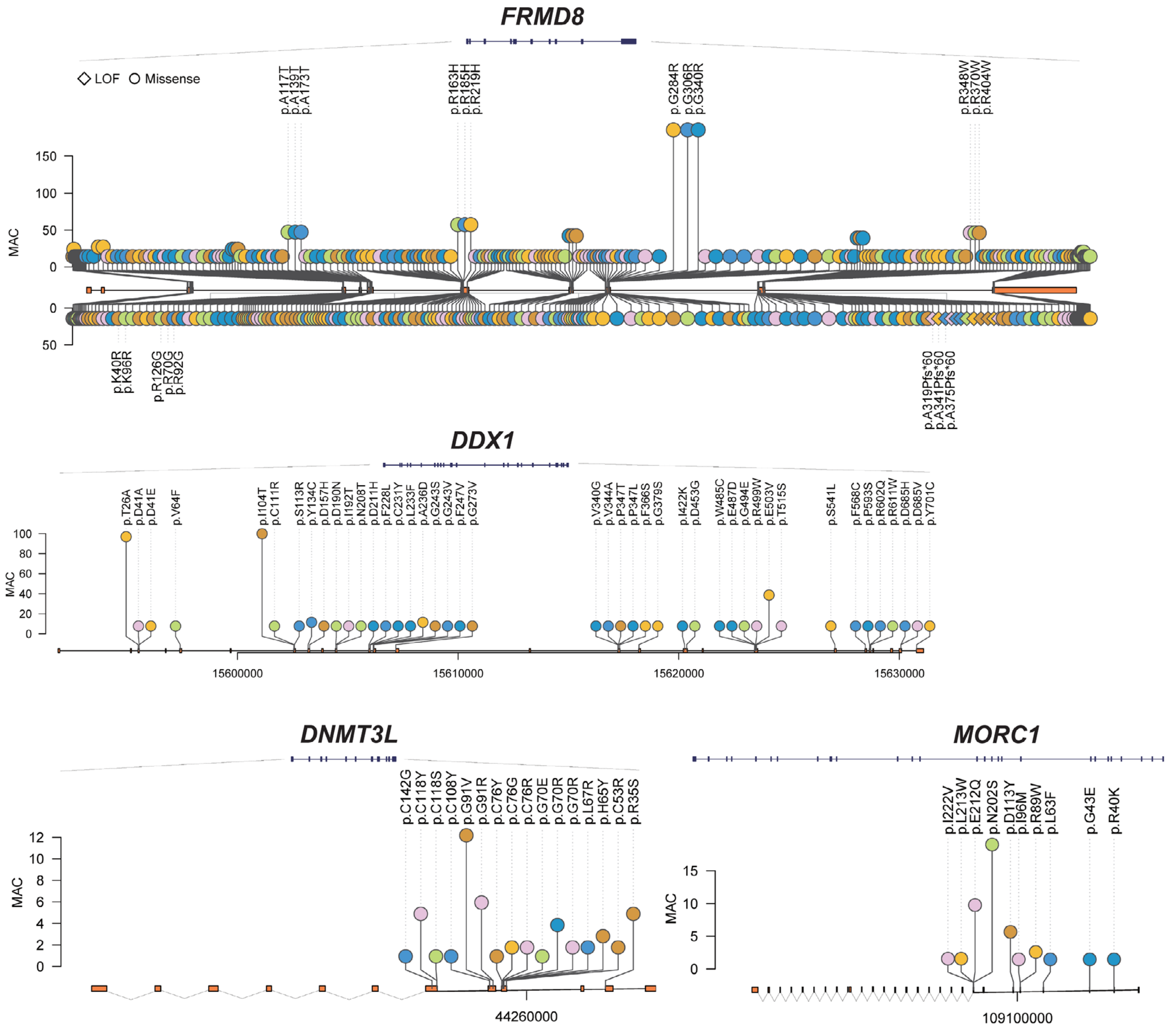

FIGURE 3 Diagram of gene structure patterns of identified novel genes. The protein-coding variants that contribute to the exome signal of the novel genes, *FRMD8*, *DDX1*, *DNMT3L*, and *MORC1*, were displayed. The height of the lollipop chart denotes the minor allele count (MAC) for each protein-coding variant.

putamen, hippocampus, and frontal cortex (Figure 5B and Table S15). Differential expression analyses based on snRNA-seq data indicated that these genes were extensively expressed in various neurons and glial cells but relatively higher in microglia (Figure 5C). Specifically, *FRMD8* and *DNMT3L* were relatively higher in cells of the vessel wall, while *DDX1* and *MORC1* were relatively higher in neurons (Figure S5). Enrichment analyses through the pre-defined gene sets of Gene Ontology revealed that these genes were overrepresented in pathways predominantly involved in the biological processes of amyloid-β (Aβ) protein formation, metabolism, and clearance; cellular components of Golgi apparatus; molecular functions of lipoprotein binding and tau binding (Figure 5A and Table S14). PPIN analyses and subsequent MCL extracted seven clusters of ADRD-associated genes. The cluster comprising *DDX1*, *MCM7*, and *WDR12* was mainly enriched in the gene set of nucleotide metabolism, and the cluster of *SLCO1C1*, *AZGP1*, *CLU*, and *CR1* was enriched in the prevention of the aggregation of non-native proteins like amyloid precursor protein (APP), lipid degradation and immune responses (Figure 5D and Table S16-S17). We also performed the enrichment analyses restricted to genes identified from ExWAS of rare variants (Table S18), and the results were similar to the main analyses. When restricted to novel genes identified in ExWAS of rare variants (Table S19), the results highlighted their critical role in regulating DNA methylation.

3.4 | The causality and predicting the performance of the identified genes in ADRD

The MR analyses integrating QTL data of brain tissues were conducted to examine the causality between our identified genes and

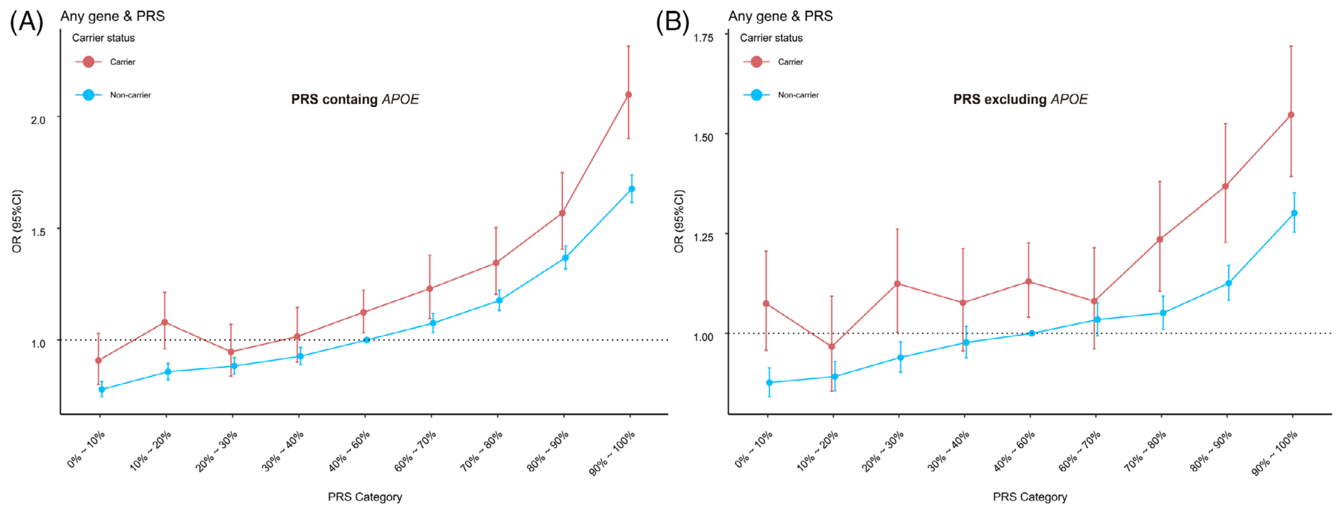

FIGURE 4 Aggregate effect of genes identified from gene-based ExWAS and PRS on ADRD risk. (A) PRS containing *APOE* region; (B) PRS excluding *APOE* region. Participants are divided into 18 groups according to the carrier status (no vs. yes) of any rare variant of the 10 identified genes, and PRS categories (0%–10%, 10%–20%, 20%–30%, 30%–40%, 40%–60%, 60%–70%, 70%–80%, 80%–90%, and 90%–100%). Logistic regression is used to investigate the combined effects with "carrier status no & PRS category 40%–60%" as the reference group.

ADRD. Using a threshold of 5×10^{-8} for instrumental variables selection, there were available eQTL data for 17 identified genes, sQTL data for eight identified genes, and pQTL data for four identified genes in brain. Expression or pre-mRNA splicing of nine genes was found to be significantly (FDR-Q < 0.05) related to ADRD (Figure 6A-C and Table S20), including the novel gene *SLCO1C1* (FDR-Q = 0.012 of sQTL-based MR). For *FRMD8* and *DNMT3L*, the MR test was unavailable due to the lack of available instrumental variables.

Next, we executed Cox regression analyses to detect the efficiency of the identified genes in predicting the incidence of ADRD. The effect sizes of genes identified from gene-based ExWAS, (highest hazard ratio [HR] and 95% CI 8.77 [5.52–13.93] for *GRN*), were substantially higher than that of genes identified from single-variant ExWAS (highest HR and 95% CI 3.06 [2.92–3.22] for *APOE*) (Figure 6D-E and Table S21-S22). Of note, individuals carrying the novel gene, *DNMT3L*, possessed an increased risk of ADRD (HR and 95% CI 4.57 [1.72–12.19], FDR-Q = 0.005). As for the novel gene *TGM2* identified from stratification analyses, we found that its predictive performance of ADRD risk was significant in the subgroup of *APOEε4* carrier (HR and 95% CI 3.19 [1.20–8.51], FDR-Q = 0.035). We further explored the interaction of *TGM2* and *APOE* in the whole population. Compared with the individuals carrying neither *TGM2* nor *APOE*, the risk for ADRD was high in the group only carrying *APOE* (HR and 95% CI 2.77 [2.67–2.87], $p < 1.0 \times 10^{-100}$), and the risk was much higher in the group carrying both *TGM2* and *APOE* (HR and 95% CI 10.39 [3.90–27.71], $p = 8.2 \times 10^{-6}$). The number of individuals only carrying *TGM2* was too small to achieve the statistical power, so we didn't test the corresponding risk. The strong additive risk effect of *TGM2* and *APOE* indicated the potential interaction of these two genes in ADRD.

3.5 | The potential of novel genes in drug development

Finally, to prioritize drug targets for ADRD, we searched the druggable genome source,[46] DrugBank database,[47] ChEMBL database,[48] and TTD[49] to find relevant evidence for identified novel genes *DDX1*, *DNMT3L*, *FRMD8*, *MORC1*, *TGM2*, *SLCO1C1*, and *NDNF*. It is indicated that five of these genes were effective targets with druggable small molecules or other types of drugs, among which there were up to 245 drugs for *DDX1*, 38 for *DNMT3L* (DNMT3A2/3L complex), 1 for *TGM2*, 20 for *SLCO1C1*, and 1 for *NDNF* (Table S23). No evidence was obtained from the databases for *FRMD8* and *MORC1*.

4 | DISCUSSION

This study leveraged large-scale exome sequencing data to elucidate the genetic determinants for ADRD. We identified 11 genes by gene-based ExWAS of rare variants and 17 genes by single-variant ExWAS of common variants. Among the identified genes, seven novel genes presented exome-wide significance for ADRD, and five of them were validated in another independent cohort, including *FRMD8*, *DDX1*, *DNMT3L*, *MORC1*, and *TGM2*. Functional annotations found that the identified genes were enriched in brain tissues and microglia as well as in pathways mainly associated with Aβ and lipid metabolism. Additional evidence from differential expression analyses, MR and Cox regression further supported the significance of these genes in ADRD. Druggability investigation of novel genes indicated that *DDX1*, *DNMT3L*, *TGM2*, *SLCO1C1*, and *NDNF* were potentially effective drug targets. These findings contribute to the current body of evidence on the genetic etiology of ADRD.

TABLE 2 Significant (FDR-adjusted) results from single-variant ExWAS of ADRD in UKB.

Gene	Variant	Chromosome	Position	Reference allele	Effect allele	Effect allele frequency	OR	SE	*p*-value	FDR-Q
APOE	chr19:44908684:T:C (rs429358)	19	44908684	T	C	0.156	1.798	0.008	<1E-100	<1E-100
PILRA	chr7:100374211:A:G (rs1859788)	7	100374211	G	A	0.319	0.945	0.007	3.E-15	2.E-11
CLU	chr8:27604964:A:G (rs7982)	8	27604964	G	A	0.404	0.951	0.007	4.E-13	1.E-09
PICALM	chr11:86014894:C:T (rs592297)	11	86014894	T	C	0.197	0.945	0.009	5.E-11	1.E-07
MS4A6A	chr11:60178272:T:C (rs12453)	11	60178272	T	C	0.405	0.958	0.007	3.E-10	6.E-07
BIN1	chr2:127106970:G:A (rs11554586)	2	127106970	G	A	0.180	1.051	0.009	7.E-09	1.E-05
APH1B	chr15:63277703:C:T (rs117618017)	15	63277703	C	T	0.140	1.052	0.010	1.E-07	2.E-04
ABI3	chr17:49219935:T:C (rs616338)	17	49219935	C	T	0.011	1.167	0.030	2.E-07	3.E-04
CR1	chr1:207505962:A:G (rs4844600)	1	207505962	G	A	0.188	1.043	0.008	5.E-07	8.E-04
ABCA7	chr19:1053525:C:G (rs3752241)	19	1053525	C	G	0.166	0.957	0.009	1.E-06	0.002
BTNL2	chr6:32396067:C:T (rs9268480)	6	32396067	C	T	0.304	0.966	0.007	2.E-06	0.002
SHARPIN	chr8:144103704:G:A (rs34173062)	8	144103704	G	A	0.076	1.059	0.012	5.E-06	0.006
SLCO1C1	chr12:20699612:C:T (rs34243130)	12	20699612	C	T	0.087	0.947	0.012	8.E-06	0.009
NDNF	chr4:121036579:T:C (rs1397645)	4	121036579	C	T	0.127	0.958	0.010	3.E-05	0.026
TMC5	chr16:19481430:C:T (rs35421097)	16	19481430	C	T	0.030	1.081	0.019	4.E-05	0.033

Abbreviations: FDR, false discovery rate; OR, odds ratio; SE, standard error.

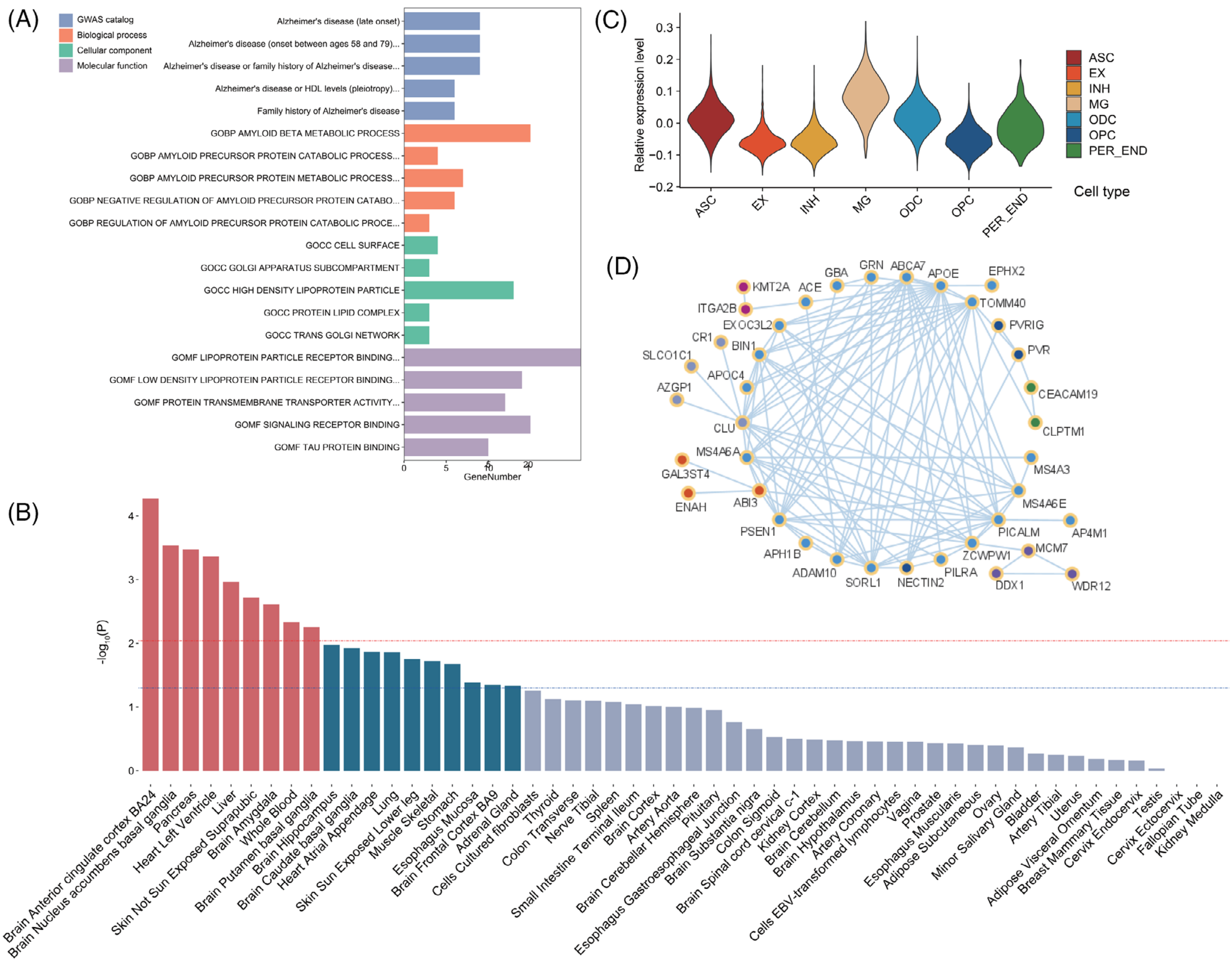

FIGURE 5 The biological function of ADRD-related genes. (A) Phenotype and GO pathway enrichment. (B) Differentially expressed gene (DEG) sets are pre-calculated by performing a two-sided t-test for any tissue type against all other 54 tissue types in GTEx v8. Tissue enrichment of ADRD-associated genes is investigated by the hypergeometric test, testing ADRD-associated genes against each of the DEG sets. The dotted blue line represents the significance level of $p = 0.05$ and the dotted red line represents the significance level of FDR-Q $= 0.05$; (C) Relative expression level of ADRD-associated genes among different cells; (D) Protein-protein interaction with at least medium confidence for ADRD associated genes are displayed followed by Markov clustering. Different colors denote different clusters. ASC, astrocyte; EX, excitatory neuron; INH, inhibitory neuron; MG, microglia; ODC, oligodendrocyte; OPC, oligodendrocyte precursor cell; PER.END pericyte and endothelial cells.

The results presented in the current study were in agreement with those reported in the literature and extended genetic research in ADRD. First, our ExWASs have overlapping findings with published GWASs. Half of the 27 ADRD-related genes identified from our study have been reported in previous large-scale GWAS of AD,[50] suggesting the convergence between ExWAS and GWAS approaches. Similarly, the pathway enrichment analysis showed consistent results with previous findings which mainly involved four function clusters including APP metabolism/Aβ formation and lipid metabolism,[51] and were supported by the interaction network analysis of the identified genes. The scRNA-seq analysis with recently released brain expression data showing genes most highly expressed in microglial cells were also in line with cell enrichment results from GWAS-identified genes,[51] highlighting their critical role in AD biological processes. Secondly, our study replicated findings from recent ExWASs. Three genes identified from our gene-based ExWAS analyses, *SORL1*, *ABCA7*, and *ADAM10*, have been reported by the ExWAS based on the ADES consortium, ADSP, and several other independent cohorts.[5] The *SORL1* was also reported in a recent paper by Wightman et al. using the UKB dataset.[52] Specifically, the significant contribution of rs140327834 (chr11:121625107:A > T) to the association for *SORL1* was also identified, which was a missense leading to aspartate-to-valine substitution and was predicted to be potentially damaging.[53,54] Compared to the work by Wightman et al. (22,080 cases and 126,428 controls, data accessed in 2020), our study used new-released genetic and phenotypic data, leading to a remarkably increased power. Additionally, we performed a detailed variant annotation with more functional categories included in the analyses, and even ultra-rare variants (MAF < 0.01%) were retained in

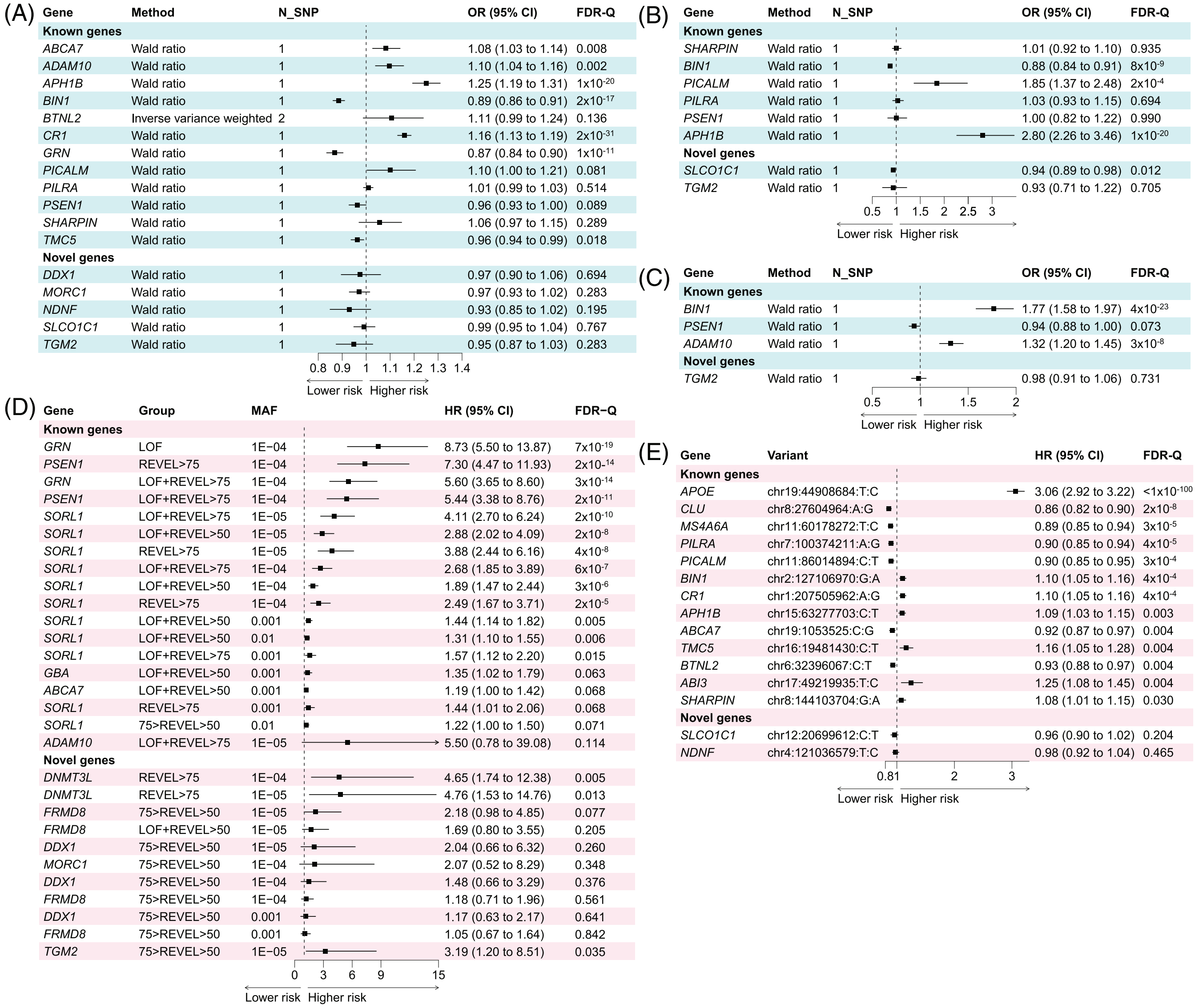

FIGURE 6 The causality and predicting performance of the identified genes in ADRD. The results of MR analyses integrating eQTL (A), sQTL (B), and pQTL (C) data in the cortex were presented as OR ± 95% CI and a t-test was utilized to obtain the two-sided *p* values. The results of Cox regression analyses of genes identified from gene-based (D) and single-variant (E) ExWAS are presented as HR ± 95% CI, and the Wald test was utilized to obtain the two-sided *p* values. FDR correction was applied to all analyses.

the statistic models, which enabled us to identify more ADRD-related genes. Importantly, a series of post-ExWAS analyses in our study were performed to enhance the robustness of findings, such as differential expression analyses, enrichment analyses, MR and Cox regression, and so on. The advantages mentioned above advance our work and promote new discoveries.

Several variants identified from the single-variant ExWAS were previously reported to be associated with AD with different underlying mechanisms. The strongest signal, *PILRA* rs1859788 (A > G), was a missense variant that led to the substitution of glycine for arginine.[55] This substitution could alter the ligand binding affinity of PILRA and has been suggested to be protective for AD.[55] Both *CLU* rs7982 and *PICALM* rs592297 are located in the 5th exon of the gene and play a functional role in gene splicing for AD. *CLU* rs7982 influences alternative splicing of *CLU* in a region-dependent and exon-specific way,[56,57] while *PICALM* rs592297is an exonic splice enhancer that impacts atrophy of the posterior cingulate and hippocampus.[58] *APH1B* rs117618017 is a missense variant in exon 1 (p.Thr27Ile) and is likely the single causal variant at its locus with a fine-mapping probability of 90%.[59] Recent studies have shown that its minor allele increases the risk of AD by directly increasing APH1B expression, rather than by changing the protein sequence.[60,61] *MS4A6A* rs12453 (T > C) has an unexpected functional outcome as well since it is a synonymous variant.[62] Its strong association with AD risk may be explained by high linkage disequilibrium with the risk variant in the CTCF binding site (rs636317).[63] While these variants have been shown to be associated with AD risk and phenotypes, the other seven major variants identified in our study including *BIN1* rs11554586, *CR1* rs4844600, *ABCA7* rs3752241, *BTNL2* rs9268480, *SLCO1C1* rs34243130, *NDNF* rs1397645, and *TMC5* rs1397645 have been rarely reported. Further investigations are needed to understand their underlying mechanism for AD.

A total of seven novel genes were identified by the ExWAS, with multiple pieces of evidence being most consistent for *FRMD8*, *DDX1*, *DNMT3L*, *MORC1*, and *TGM2*. *FRMD8* (FERM domain containing eight) encodes a protein that promotes the cell surface stability and regulates the inflammatory and growth factor signaling.[64,65] *DDX1* (DEAD-box helicase 1) encodes protein characterized by the conserved motif Asp-Glu-Ala-Asp (DEAD) that acts as an ATP-dependent RNA helicase. Despite moderate-to-strong evidence from our differential expression analyses in the brains of AD versus controls, the biological functions of *FRMD8* and *DDX1* in the central nervous system remained largely undiscovered. Future experimental research is needed to clarify their functional relevance to AD as well as their causal molecular mechanisms. *DNMT3L* (DNA methyltransferase-3-like) encodes an enzymatically inactive regulatory factor of DNA methyltransferases that can either promote or inhibit DNA methylation depending on the context.[66] Recent studies found that the interaction of nutrition and genetics via *DNMT3L*-mediated DNA methylation determined cognitive decline[67] and *DNMT3L* had differentially hydroxymethylated regions in AD *post mortem* brain tissues versus controls.[68] This supported our findings and suggested that the *DNMT3L* was involved in the pathophysiology of AD. The protein encoded by *MORC1* (Microchidia family CW-type zinc finger 1) is required for spermatogenesis and essential for de novo DNA methylation and silencing of transposable elements in the male embryonic germ cells. Previous research found its methylation was associated with microstructural features of the hippocampus and medial prefrontal cortex, and it was suggested to be involved in the neurobiological process of neuropsychiatric disorders.[69,70] *TGM2* (Transglutaminase 2) encodes a multifunctional transamidation acyltransferase that catalyzes calcium-dependent post-translational modifications that induce protein cross-linking, glutamine deamination, polyamine incorporation, and guanosine triphosphate hydrolysis.[71,72] A recent experimental study found an association between *TGM2* and Alzheimer's pathogenesis. The silencing of *TGM2* could prevent Aβ-induced mitochondrial calcium influx, ROS accumulation, Tau phosphorylation, and cell death in neuronal cells.[71] This result may support one of our interesting findings that rare *TGM2* variants in *APOE4* carriers were associated with a further increased risk of incident ADRD.

The study of rare variants promotes a full understanding of the genetic architecture for AD. With both exome sequencing and genome-wide genotype data in the UKB, we investigated the relative contribution of common variant-based PRS and rare coding variant burden to AD risk. Carriers of such variants had higher risks for AD and incremental risk showed an increasing trend with PRS, which implied that AD risk might vary depending on an individual's genetic profile of common risk variants and carrier status for these rare variants and rare coding variants affected additively to PRS. This suggests that carrying rare or common risk variants could serve as distinct pathways to AD onset and that genetic prediction may be further improved by combining PRS and the rare coding variant burden. Similar findings were reported previously for other common complex phenotypes.[73,74]

Intriguingly, our study provides some genetic support for the sex difference in AD. In stratification analyses of gene-based ExWAS, genes such as *SORL1* and *PSEN1* showed significant associations with AD in females whereas no genes were identified in males. Additionally, the variant rs7982 (*CLU*) identified in the single variant ExWAS was previously found to be associated with alternative splicing of CLU in all three brain regions tested in women including the temporal cortex, superior temporal gyrus, and parahippocampal gyrus, while the association was significant only in temporal cortex in men.[56] These results may provide a clue for further research into the sex's effect on AD pathogenesis.

Our study had several limitations. First, the study was conducted among White British participants. It is undetermined how well the results from our study can be extrapolated to other ancestries. We anticipate that large, diverse samples will become more readily available to further verify our findings and make new discoveries. Second, proxy ADRD and non-AD dementia cases were included. Although the sample size and power can be boosted with this approach, the discovered genes should be cautiously interpreted as phenotypic heterogeneity may point to dementia-related genes instead of AD-specific genes. Further analysis in AD cohorts or experimental research is also warranted to validate the findings from the study with a similar design. In particular, the functional consequences of the identified rare variants in relation to AD development need to be further investigated. Third, our study did not assess the effect of other structural variants such as copy number variants and repetitive sequences. Evaluations of these genetic variations will contribute to the complete characterization of the genetic background of AD and may provide additional insights into the disease mechanism. Fourth, The FinnGen data replication were based on single variants, thus a strict replication analysis of the burden of rare variants cannot be performed in this cohort. Finally, even if the coding variants in gene-based analyses are rare, it cannot be excluded that they still tag a haplotype that has another variant(s).

In conclusion, our study using WES data from a large cohort identified novel genes and variants for ADRD and confirmed the well-established genetic factors. The findings strengthen the current understanding of the genetic architecture of AD and provide new insights into pathological mechanisms and therapeutic targets.

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CONFLICT OF INTEREST STATEMENT

Authors declare that they have no competing interests. Author disclosures are available in the supporting information.

CONSENT STATEMENT

All participants provided written informed consent according to the Declaration of Helsinki before study enrollment.

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